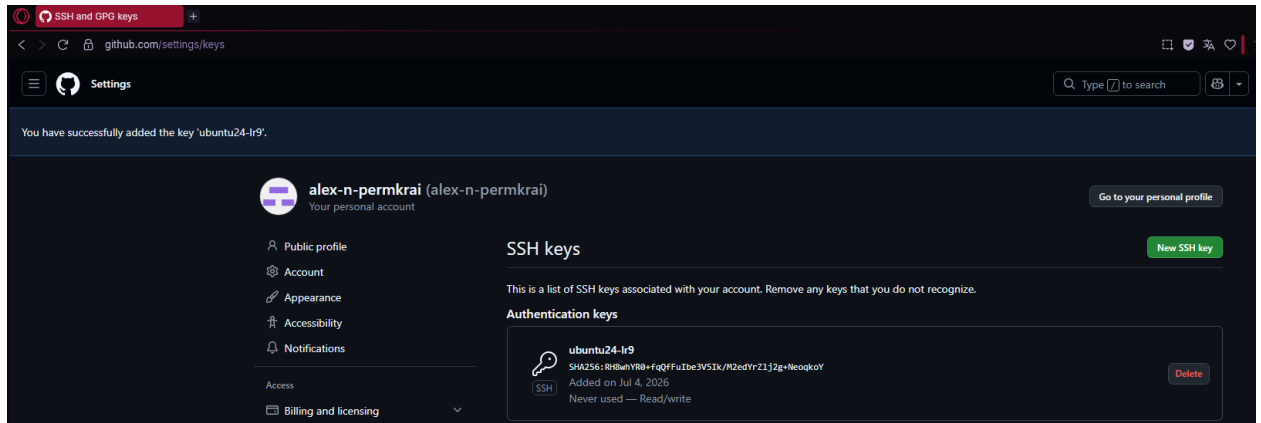
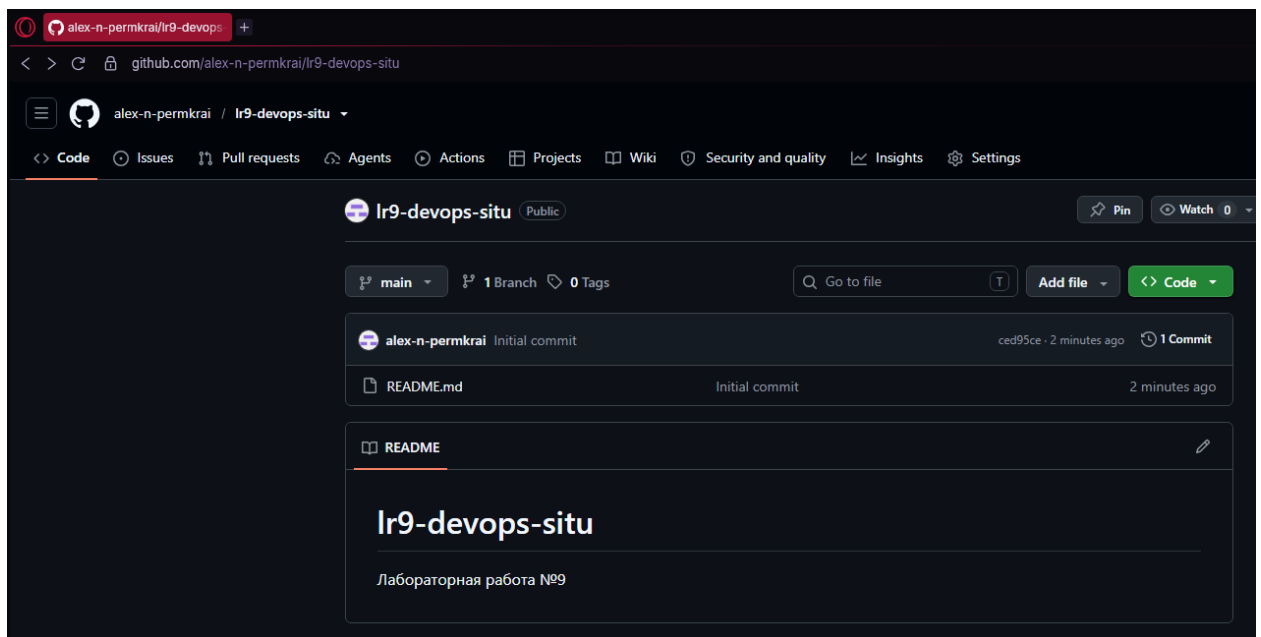


Добавляем на github



```
root@ubuntu-1:/tmp# ssh -T git@github.com
The authenticity of host 'github.com (140.82.112.4)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvC0qU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi alex-n-permkrai! You've successfully authenticated, but GitHub does not provide shell access.
root@ubuntu-1:/tmp#
```

3. Создание репозитория GitHub



```
root@ubuntu-1:~/lr9-devops-situ# git config user.name "alex-n-permkrai"
root@ubuntu-1:~/lr9-devops-situ# git config user.email "alex-n-permkrai@users.noreply.github.com"
root@ubuntu-1:~/lr9-devops-situ# git switch -c dev
Switched to a new branch 'dev'
root@ubuntu-1:~/lr9-devops-situ# git branch
* dev
  main
root@ubuntu-1:~/lr9-devops-situ#
```

4. Создаём приложение

Создаем каталоги, requirements.txt, server/application.py, server/test_application.py, server/index.html, server/dockerfile

5. Локальная проверка приложения, тестов и Docker

```
Successfully built 89efa30cf38d
Successfully tagged test-image:latest
(.venv) root@ubuntu-1:~/lr9-devops-situ# docker rm -f app-server8000 2>/dev/null || true
docker run -d --name app-server8000 -p 8000:8000 test-image
b3cf39e32be73ea186dff86c58506231a1aeda3a77bde7621f078f362d68de76
(.venv) root@ubuntu-1:~/lr9-devops-situ# docker ps
curl http://127.0.0.1:8000
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
b3cf39e32be7   test-image  "python3 -u /usr/loc..." 6 seconds ago  Up 5 seconds  0.0.0.0:8000->8000/tcp           app-server8000
Hello, v1
```

6. GitHub Actions CI

Создаем .github/workflows/cicd.yml

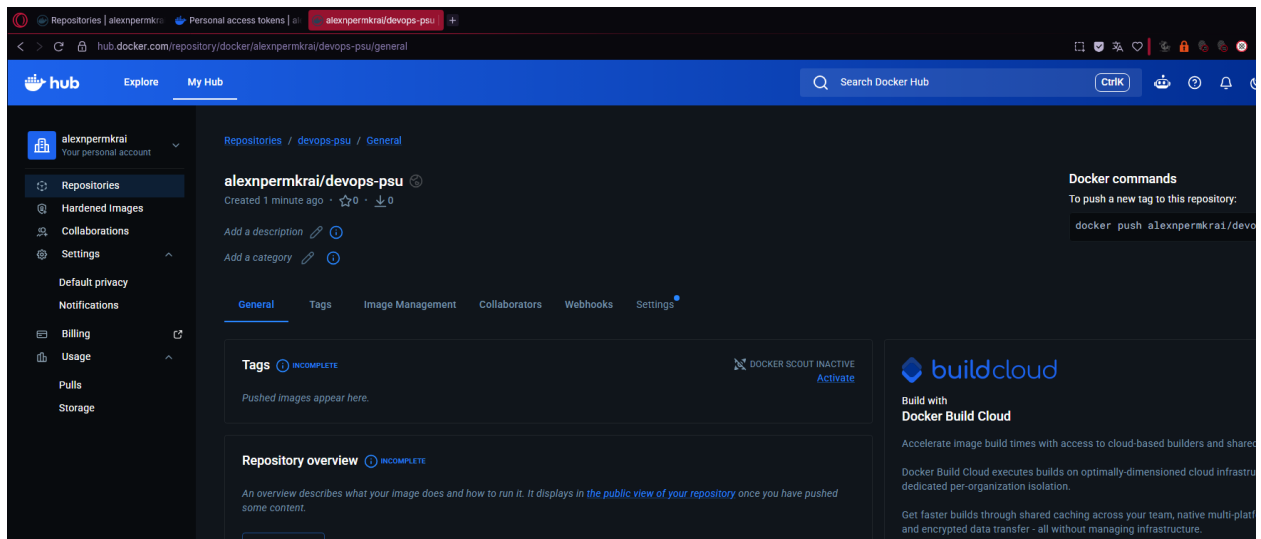
7. K8s manifest

Создаем server-k8s-manifests/devops-psu.yml

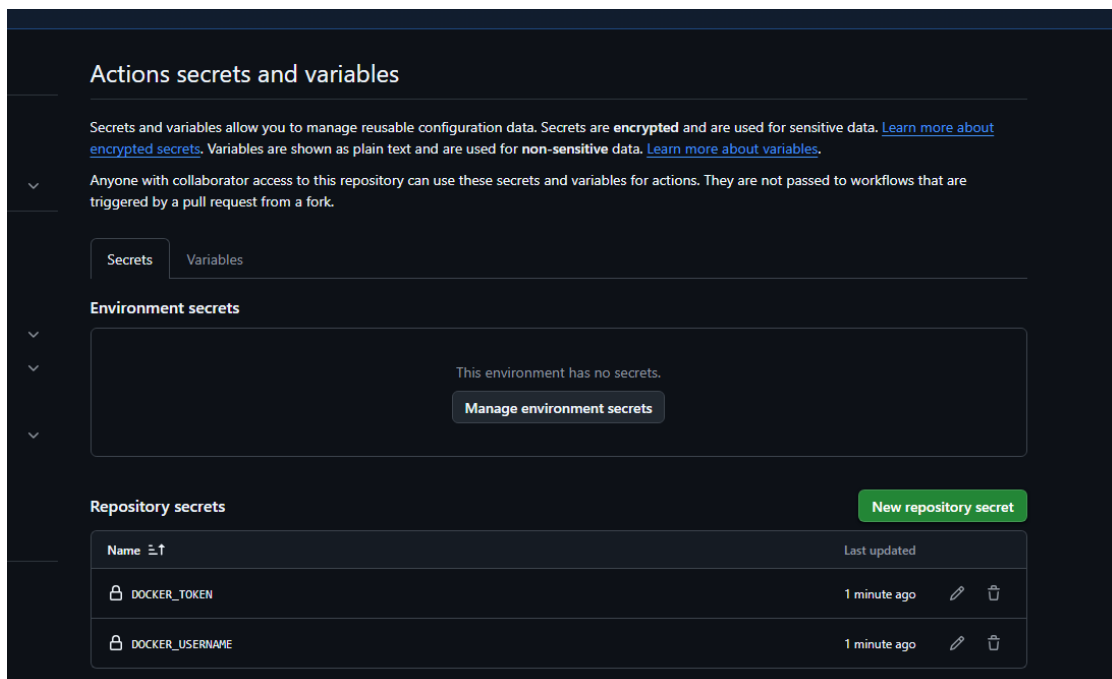
DockerHub-образ будет называться: alexnpermkrai/devops-psu:latest

8. GitHub Actions CD: build and push DockerHub

Создаем на DockerHub публичный репозиторий alexnpermkrai/devops-psu



DockerHub Access Token добавляем в секреты на гитхаб



Создаем .github/workflows/release.yml

9. Коммит в dev

```
root@ubuntu-1:~/lr9-devops-situ# tree -a -I ".git|.venv"
.
├── .github
│   └── workflows
│       ├── cicd.yml
│       └── release.yml
├── .pytest_cache
│   ├── CACHEDIR.TAG
│   ├── .gitignore
│   ├── README.md
│   └── v
│       └── cache
│           └── nodeids
├── README.md
├── requirements.txt
├── server
│   ├── application.py
│   ├── dockerfile
│   ├── index.html
│   ├── __pycache__
│   │   ├── application.cpython-312.pyc
│   │   └── test_application.cpython-312-pytest-9.1.1.pyc
│   ├── test_application.py
├── server-k8s-manifests
│   └── devops-psu.yml
└── 9 directories, 15 files
root@ubuntu-1:~/lr9-devops-situ#
```

```
root@ubuntu-1:~/lr9-devops-situ# git push -u origin dev
Enumerating objects: 2818, done.
Counting objects: 100% (2818/2818), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2768/2768), done.
Writing objects: 100% (2817/2817), 10.27 MiB | 4.32 MiB/s, done.
Total 2817 (delta 168), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (168/168), done.
remote:
remote: Create a pull request for 'dev' on GitHub by visiting:
remote:   https://github.com/alex-n-permkrai/lr9-devops-situ/pull/new/dev
remote:
To github.com:alex-n-permkrai/lr9-devops-situ.git
 * [new branch]      dev -> dev
branch 'dev' set up to track 'origin/dev'.
root@ubuntu-1:~/lr9-devops-situ#
```

10. Проверка CI в GitHub Actions

The screenshot shows the GitHub Actions interface for the repository 'alex-n-permkrai / lr9-devops-situ'. The 'Actions' tab is selected, displaying a list of workflows. The workflow 'Initial cloud-native app with CI CD manifests' is shown as 'In progress' on the 'dev' branch, triggered by a commit '904c2d5' pushed by 'alex-n-permkrai' 1 minute ago. The workflow run details show a step 'LINT-TEST-BUILD-CHECK #1'.

Workflow runs: alex-n-permkrai

github.com/alex-n-permkrai/lr9-devops-situ/actions

alex-n-permkrai / lr9-devops-situ

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Actions

New workflow

All workflows

Showing runs from all workflows

1 workflow run

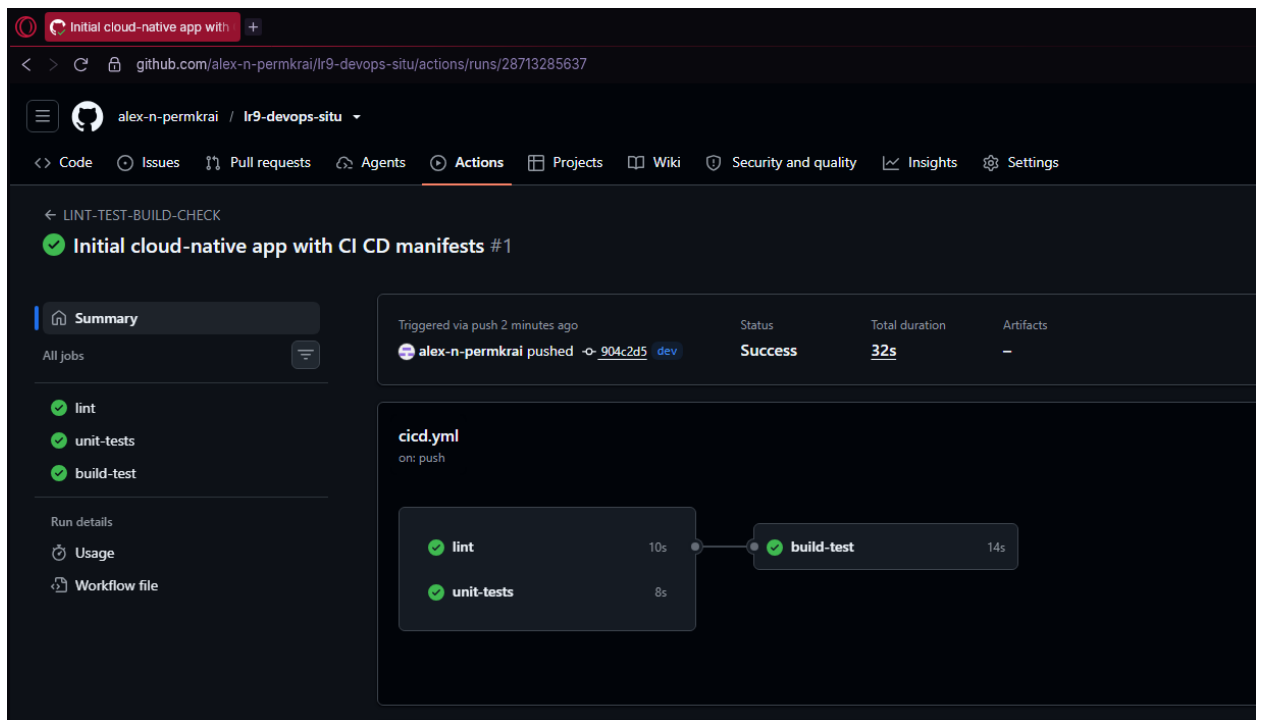
Initial cloud-native app with CI CD manifests

LINT-TEST-BUILD-CHECK #1: Commit 904c2d5 pushed by alex-n-permkrai

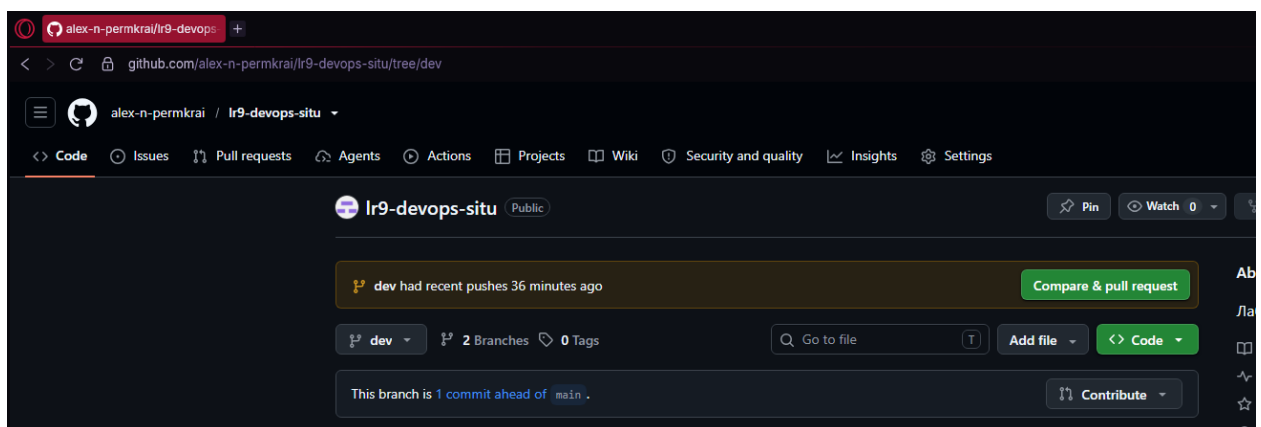
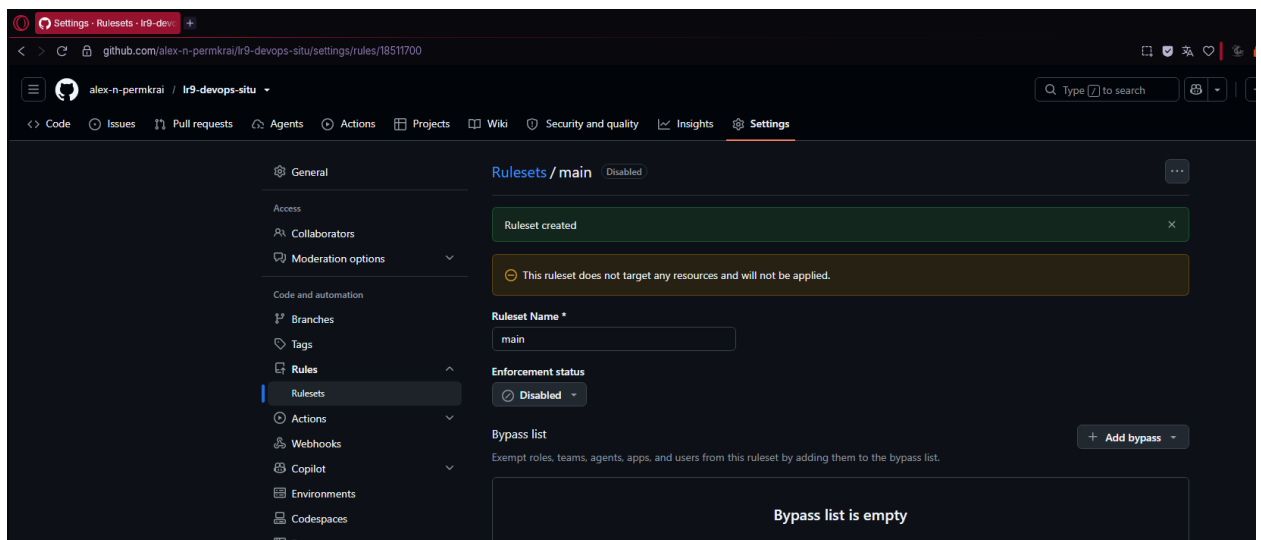
dev

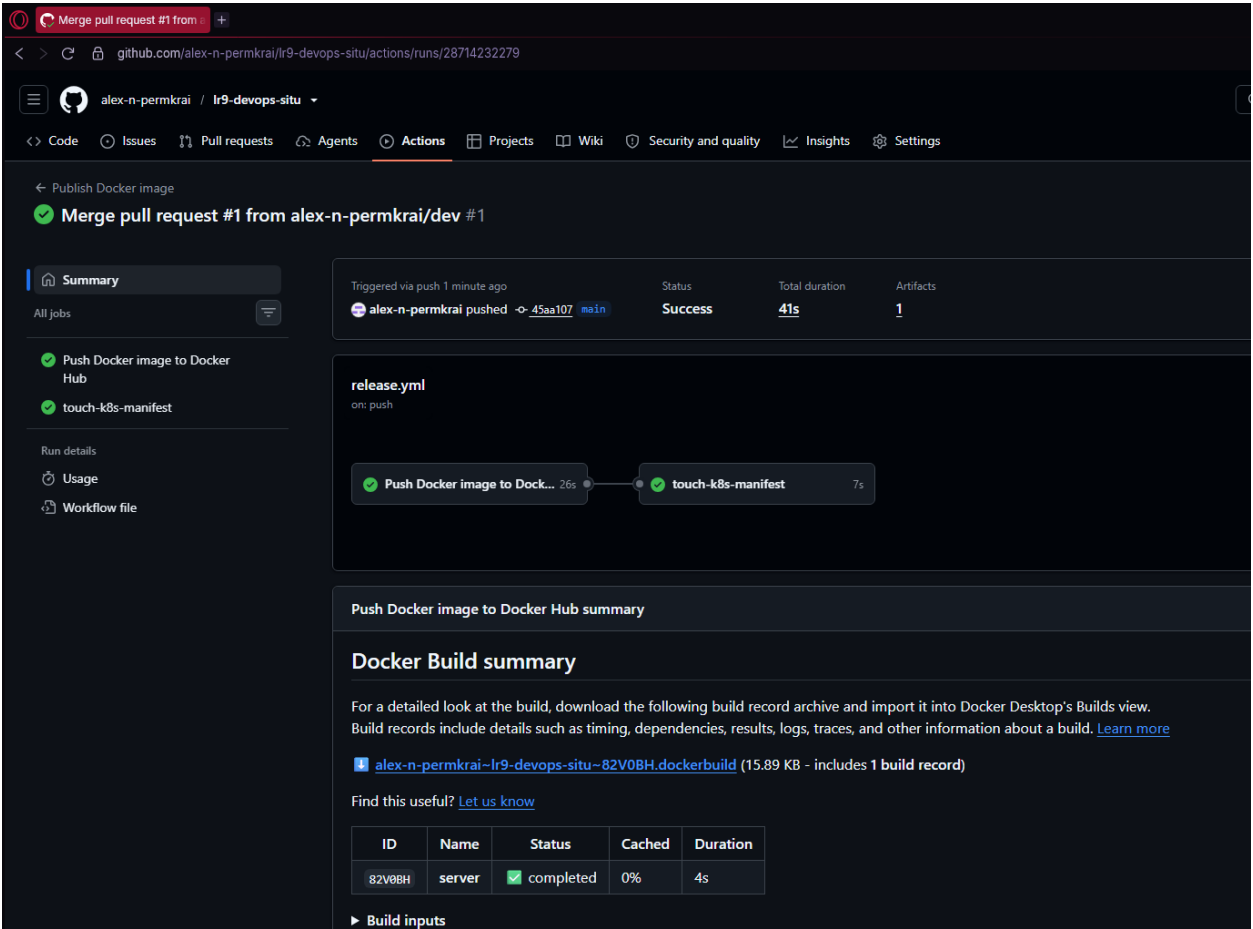
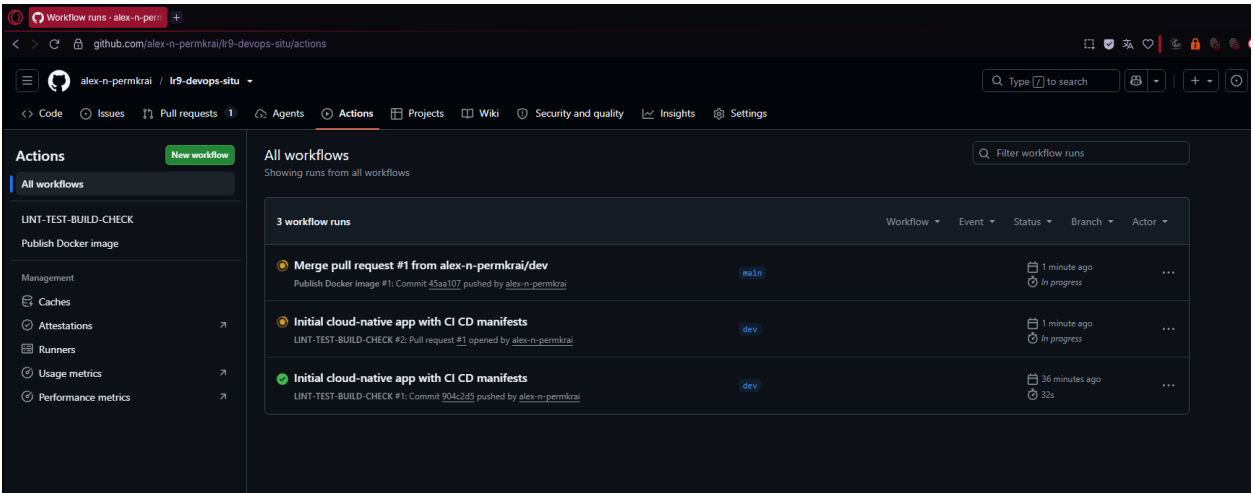
1 minute ago

In progress

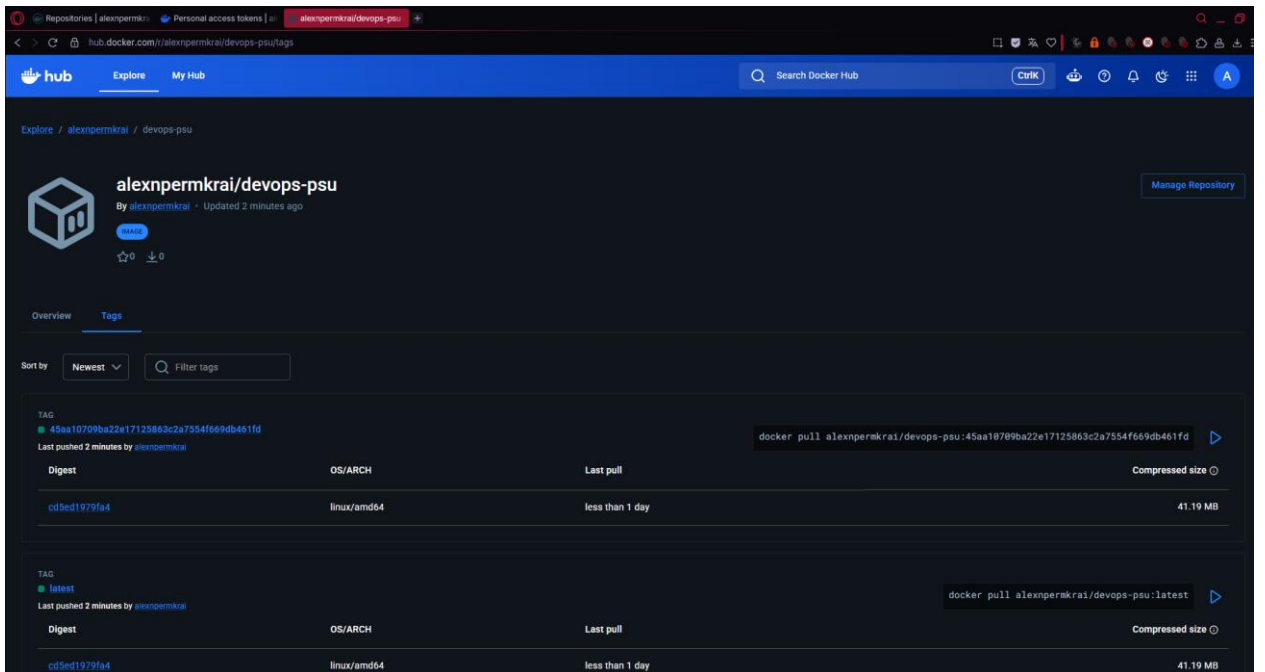
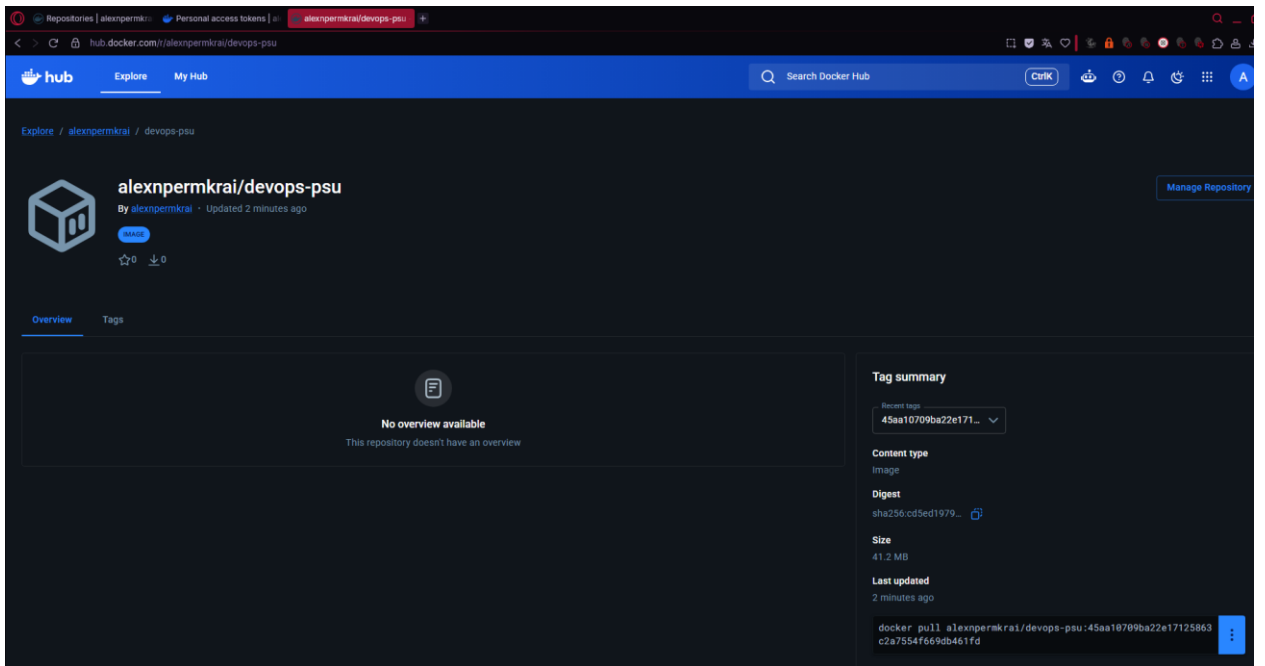


11. Защита ветки main



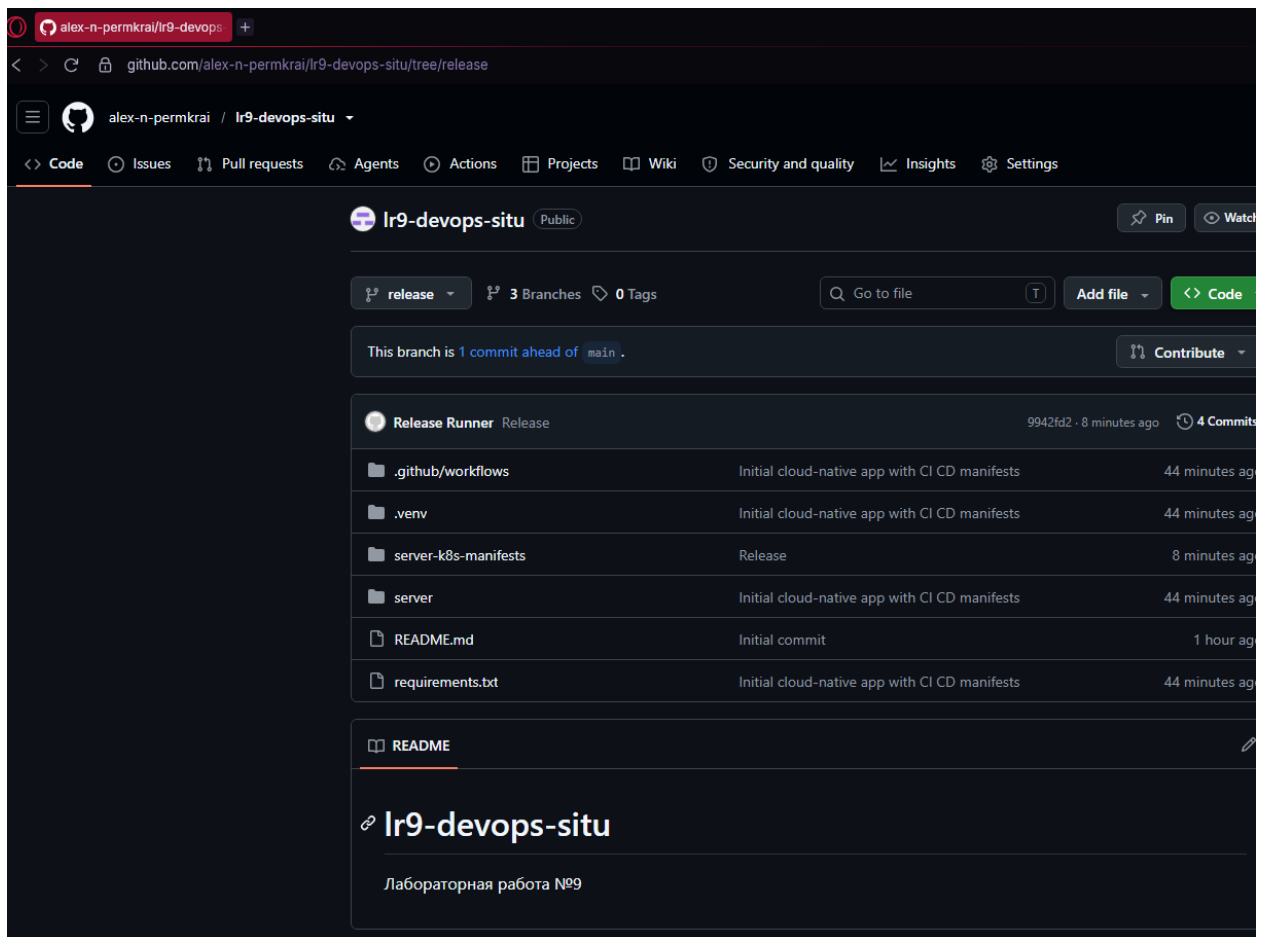


12. Проверка DockerHub



13. Проверка ветки release

```
root@ubuntu-1:~/lr9-devops-situ# git branch -a
* dev
  main
remotes/origin/HEAD -> origin/main
remotes/origin/dev
remotes/origin/main
remotes/origin/release
root@ubuntu-1:~/lr9-devops-situ# git switch release
branch 'release' set up to track 'origin/release'.
Switched to a new branch 'release'
root@ubuntu-1:~/lr9-devops-situ# cat server-k8s-manifests/devops-psu.yml
apiVersion: v1
kind: Namespace
metadata:
  name: devops-psu
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: devops-psu
  namespace: devops-psu
  labels:
    app: devops-psu
    release-date: "1783186555"
spec:
  replicas: 1
  selector:
    matchLabels:
      app: devops-psu
  template:
    metadata:
      labels:
        app: devops-psu
        svc: frontend
        release-date: "1783186555"
    spec:
      containers:
        - name: devops-psu-server
          image: alexnpermkrai/devops-psu:latest
          imagePullPolicy: Always
          ports:
            - containerPort: 8000
---
apiVersion: v1
kind: Service
metadata:
  name: service-devops
  namespace: devops-psu
  labels:
    app: devops-psu
spec:
  type: LoadBalancer
  selector:
```

14. Проверка Kubernetes-манифеста вручную

```
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-124-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of C6 04 июл 2026 17:57:11 UTC

System load:  0.34           Processes:            258
Usage of /:   57.1% of 17.54GB Users logged in:          2
Memory usage: 15%           IPv4 address for enp0s3: 10.0.0.100
Swap usage:   0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
  just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Расширенное поддержание безопасности (ESM) для Applications выключено.

9 обновлений может быть применено немедленно.
Чтобы просмотреть дополнительные обновления выполните: apt list --upgradable

4 дополнительных обновления безопасности могут быть применены с помощью ESM Apps.
Подробнее о включении службы ESM Apps at https://ubuntu.com/esm

*** Требуется перезагрузка системы ***
Last login: Sat Jul  4 16:21:09 2026 from 10.0.0.218
user@ubuntu-1:~$ sudo su
[sudo] password for user:
root@ubuntu-1:/home/user# minikube tunnel
Status:
  machine: minikube
  pid: 71077
  route: 10.96.0.0/12 -> 192.168.49.2
  minikube: Running
  services: [service-devops]
  errors:
    minikube: no errors
    router: no errors
    loadbalancer emulator: no errors
```

```

root@ubuntu-1:~/lr9-devops-situ# minikube status
minikube
type: Control Plane
host: Running
kubelnet: Running
apiserver: Running
kubeconfig: Configured

root@ubuntu-1:~/lr9-devops-situ# kubectl get nodes
NAME      STATUS   ROLES    AGE     VERSION
minikube  Ready    control-plane  4m50s   v1.35.1
root@ubuntu-1:~/lr9-devops-situ# git switch release
Already on 'release'
Your branch is up to date with 'origin/release'.
root@ubuntu-1:~/lr9-devops-situ# kubectl apply -f server-k8s-manifests/devops-psu.yml
namespace/devops-psu created
deployment.apps/devops-psu created
service/service-devops created
root@ubuntu-1:~/lr9-devops-situ# kubectl get ns
NAME      STATUS   AGE
default   Active   5m4s
devops-psu  Active   5s
kube-node-lease  Active   5m4s
kube-public  Active   5m4s
kube-system  Active   5m4s
root@ubuntu-1:~/lr9-devops-situ# kubectl get all -n devops-psu
NAME                                READY   STATUS            RESTARTS   AGE
pod/devops-psu-6d6cf675b4-86rdk     0/1     ContainerCreating  0           14s

NAME                TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
service/service-devops  LoadBalancer  10.97.141.170  <pending>      12345:30526/TCP  14s

NAME                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/devops-psu  0/1     1             0           14s

NAME                DESIRED   CURRENT   READY   AGE
replicaset.apps/devops-psu-6d6cf675b4  1         1         0       14s
root@ubuntu-1:~/lr9-devops-situ# kubectl wait --for=condition=ready pod -l app=devops-psu -n devops-psu --timeout=180s
pod/devops-psu-6d6cf675b4-86rdk condition met
root@ubuntu-1:~/lr9-devops-situ# kubectl get pods -n devops-psu -o wide
NAME                READY   STATUS    RESTARTS   AGE   IP           NODE    NOMINATED NODE   READINESS GATES
devops-psu-6d6cf675b4-86rdk  1/1     Running   0           27s   10.244.0.3   minikube  <none>           <none>

root@ubuntu-1:~/lr9-devops-situ# kubectl get svc -n devops-psu
NAME                TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
service-devops      LoadBalancer  10.97.141.170  10.97.141.170  12345:30526/TCP  114s
root@ubuntu-1:~/lr9-devops-situ# curl http://10.97.141.170:12345
Hello, v1
root@ubuntu-1:~/lr9-devops-situ#

```

15. Установка ArgoCD

```

root@ubuntu-1:~/lr9-devops-situ# kubectl get all -n argocd
NAME                                READY   STATUS            RESTARTS   AGE
pod/argocd-application-controller-0  0/1     ContainerCreating  0           26s
pod/argocd-applicationset-controller-77497b89df-rkgw5  0/1     ContainerCreating  0           27s
pod/argocd-dex-server-7c874c5958-c5hs4  0/1     Init:0/1          0           27s
pod/argocd-notifications-controller-6c5f7c5dcc-rq2xl  0/1     ContainerCreating  0           26s
pod/argocd-repo-server-59d57b7dcc-rrxlx  0/1     Init:0/1          0           26s
pod/argocd-server-7c8986577c-r9h7d  0/1     ContainerCreating  0           26s

NAME                TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
service/argocd-applicationset-controller  ClusterIP  10.109.90.200  <none>         7000/TCP,8080/TCP  27s
service/argocd-dex-server  ClusterIP  10.100.185.235  <none>         5556/TCP,5557/TCP,5558/TCP  27s
service/argocd-metrics  ClusterIP  10.101.231.184  <none>         8082/TCP          27s
service/argocd-notifications-controller-metrics  ClusterIP  10.97.233.110  <none>         9001/TCP          27s
service/argocd-repo-server  ClusterIP  10.107.69.233  <none>         6379/TCP          27s
service/argocd-server  ClusterIP  10.102.30.197  <none>         8081/TCP,8084/TCP  27s
service/argocd-server-metrics  ClusterIP  10.101.231.132  <none>         80/TCP,443/TCP    27s
service/argocd-server-metrics  ClusterIP  10.106.61.167  <none>         8083/TCP          27s

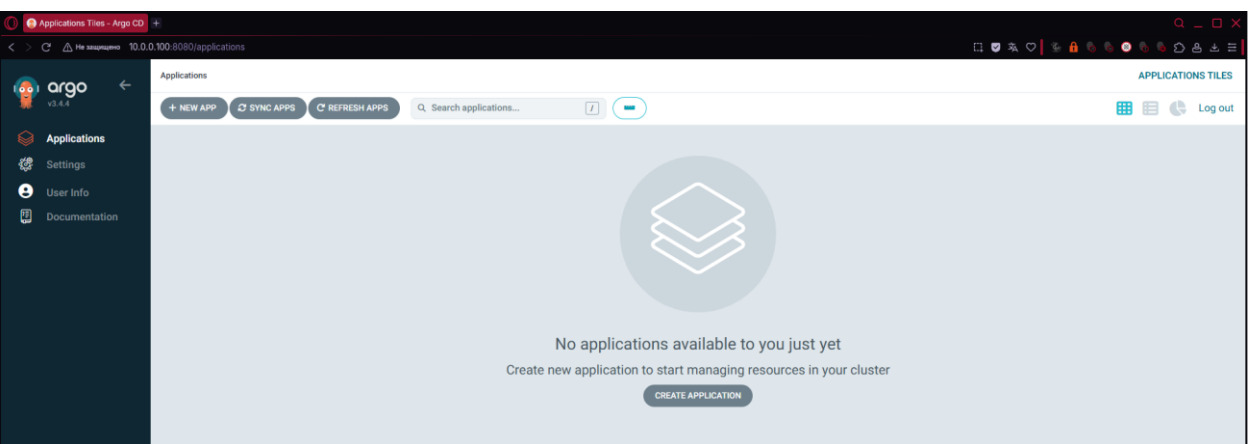
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/argocd-applicationset-controller  0/1     1             0           27s
deployment.apps/argocd-dex-server  0/1     1             0           27s
deployment.apps/argocd-notifications-controller  0/1     1             0           27s
deployment.apps/argocd-repo-server  0/1     1             0           26s
deployment.apps/argocd-server  0/1     1             0           26s

NAME                DESIRED   CURRENT   READY   AGE
replicaset.apps/argocd-applicationset-controller-77497b89df  1         1         0       27s
replicaset.apps/argocd-dex-server-7c874c5958  1         1         0       27s
replicaset.apps/argocd-notifications-controller-6c5f7c5dcc  1         1         0       26s
replicaset.apps/argocd-repo-server-59d57b7dcc  1         1         0       26s
replicaset.apps/argocd-server-7c8986577c  1         1         0       26s

NAME                READY   AGE
statefulset.apps/argocd-application-controller  0/1     26s
root@ubuntu-1:~/lr9-devops-situ# kubectl wait --for=condition=available deployment --all -n argocd --timeout=300s
deployment.apps/argocd-applicationset-controller condition met
deployment.apps/argocd-dex-server condition met
deployment.apps/argocd-notifications-controller condition met
deployment.apps/argocd-repo-server condition met
deployment.apps/argocd-server condition met
root@ubuntu-1:~/lr9-devops-situ#
root@ubuntu-1:~/lr9-devops-situ# kubectl get pods -n argocd
NAME                                READY   STATUS    RESTARTS   AGE
argocd-application-controller-0     1/1     Running   0           3m33s
argocd-applicationset-controller-77497b89df-rkgw5  1/1     Running   0           3m34s
argocd-dex-server-7c874c5958-c5hs4  1/1     Running   0           3m34s
argocd-notifications-controller-6c5f7c5dcc-rq2xl  1/1     Running   0           3m33s
argocd-repo-server-59d57b7dcc-rrxlx  1/1     Running   0           3m33s
argocd-server-7c8986577c-r9h7d  1/1     Running   0           3m33s
root@ubuntu-1:~/lr9-devops-situ#

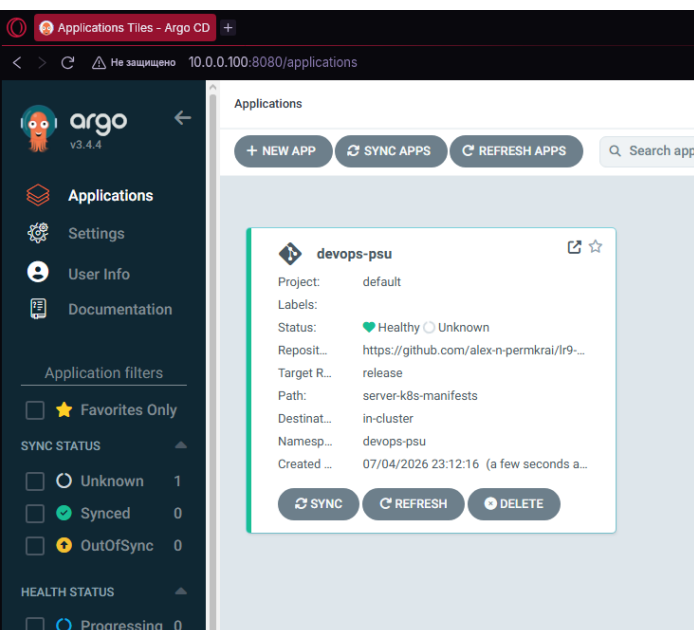
```

16. Вход в ArgoCD UI

[illegible]

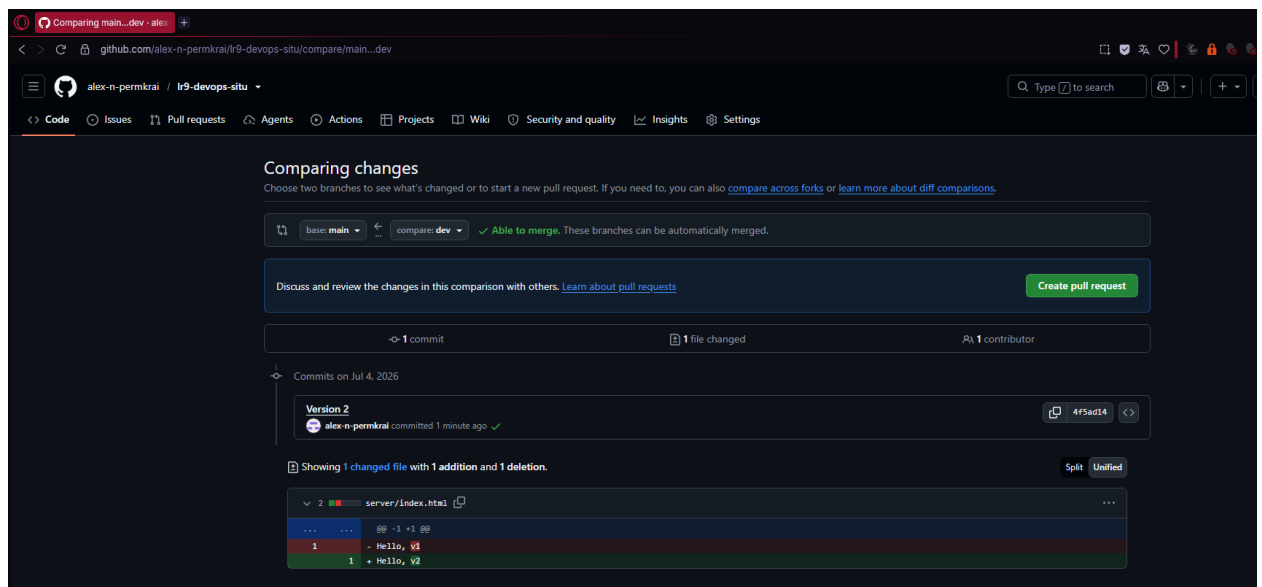
17. Подключение приложения в ArgoCD

```
root@ubuntu-1:~/lr9-devops-situ# kubectl apply -f argocd/application.yml
application.argoproj.io/devops-psu created
root@ubuntu-1:~/lr9-devops-situ# kubectl get application -n argocd
NAME          SYNC STATUS   HEALTH STATUS
devops-psu
root@ubuntu-1:~/lr9-devops-situ#
```



18. Проверка полного CI/CD workflow: версия v2

```
root@ubuntu-1:~/lr9-devops-situ# cat > server/index.html <<'EOF'
Hello, v2
EOF
root@ubuntu-1:~/lr9-devops-situ# cat server/index.html
Hello, v2
root@ubuntu-1:~/lr9-devops-situ# git add server/index.html
root@ubuntu-1:~/lr9-devops-situ# git commit -m "Version 2"
[dev 4f5ad14] Version 2
 1 file changed, 1 insertion(+), 1 deletion(-)
root@ubuntu-1:~/lr9-devops-situ# git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 334 bytes | 334.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:alex-n-permkrai/lr9-devops-situ.git
   904c2d5..4f5ad14  dev -> dev
root@ubuntu-1:~/lr9-devops-situ#
```



После merge:

1. Запустится Publish Docker image.
2. Новый образ уйдёт в DockerHub.
3. Workflow изменит release-date.
4. Обновит ветку release.
5. ArgoCD увидит изменение.
6. ArgoCD перезапустит pod.
7. Kubernetes подтянет новый latest.

```

version: v1
Kind: Service
Name: service-devops
Namespace: devops-psu
Status: Synced
Version: v1
Group: apps
Kind: Deployment
Name: devops-psu
Namespace: devops-psu
Status: Synced
Version: v1
Source Hydrator:
Source Type: Directory
Summary:
Images:
  alexnpermkrai/devops-psu:latest
Sync:
  Compared To:
    Destination:
      Namespace: devops-psu
      Server: https://kubernetes.default.svc
    Source:
      Path: server-k8s-manifests
      Repo URL: https://github.com/alex-n-permkrai/lr9-devops-situ.git
      Target Revision: release
      Revision: 2e9cd238fd080601a65be741c20c66a8023b8c6c
      Status: Synced
Events:
  Type      Reason          Age   From                                     Message
  ----      -
Normal      ResourceUpdated  16m   argocd-application-controller           Updated sync status: -> Unknown
Normal      ResourceUpdated  16m   argocd-application-controller           Updated health status: -> Healthy
Normal      OperationStarted 80s   argocd-application-controller           Initiated automated sync to '2e9cd238fd080601a65be741c20c66a8023b8c6c'
Normal      ResourceUpdated  80s   argocd-application-controller           Updated sync status: Unknown -> OutOfSync
Normal      OperationCompleted 80s   argocd-application-controller           Sync operation to 2e9cd238fd080601a65be741c20c66a8023b8c6c succeeded
Normal      ResourceUpdated  80s   argocd-application-controller           Updated sync status: OutOfSync -> Synced
Normal      ResourceUpdated  80s   argocd-application-controller           Updated health status: Healthy -> Progressing
root@ubuntu-1:~/lr9-devops-situ# curl http://10.97.141.170:12345
Hello, v2
root@ubuntu-1:~/lr9-devops-situ#

```

Workflow runs - alex-n-permkrai

alex-n-permkrai / lr9-devops-situ

Code Issues Pull requests 1 Agents Actions Projects Wiki Security and quality Insights Settings

Actions New workflow

All workflows Showing runs from all workflows

9 workflow runs

Workflow	Event	Status	Branch	Actor
Merge pull request #3 from alex-n-permkrai/release	main	In progress	main	alex-n-permkrai
Release	release	In progress	release	alex-n-permkrai
Remove virtual environment from repository	dev	Completed	dev	alex-n-permkrai
Merge pull request #2 from alex-n-permkrai/dev	main	Completed	main	alex-n-permkrai
Version 2	dev	Completed	dev	alex-n-permkrai
Version 2	dev	Completed	dev	alex-n-permkrai
Merge pull request #1 from alex-n-permkrai/dev	main	Completed	main	alex-n-permkrai
Initial cloud-native app with CI CD manifests	dev	Completed	dev	alex-n-permkrai

devops-psu - Application Details

10.0.0.100:8080/applications/argocd/devops-psu

Argo v3.4.4

Applications Settings User Info Documentation

Resource filters

NAME

KINDS

SYNC STATUS

HEALTH STATUS

DETAILS DIFF SYNC SYNC STATUS HISTORY AND ROLLBACK DELETE REFRESH

APP HEALTH: Progressing

SYNC STATUS: Synced to release (2e9cd23)

LAST SYNC: Sync OK to 2e9cd23

Diagram showing resources: devops-psu, service-devops, devops-psu, devops-psu-7498b8d985, devops-psu-6d5cf75b4, devops-psu-7498b8d985-b7zbf.

argo

v3.4.4

Applications

Settings

User Info

Documentation

Resource filters

NAME

NAME

KINDS

KINDS

SYNC STATUS

☐ Synced 3

☐ OutOfSync 0

HEALTH STATUS

☐ Progressing 0

☐ Suspended 0

☒ Healthy 5

☐ Degraded 0

devops-psu - Application

10.0.0.100:8080/applications/argo/devops-psu

APPLICATION DETAILS

DETAILS

DIFF

SYNC

SYNC STATUS

HISTORY AND ROLLBACK

DELETE

REFRESH

APP HEALTH

Healthy

SYNC STATUS

Synced to release (2e9cd23)

Auto sync is enabled.

Author: alex-n-permikai-alex-n-permikai@users.noreply.github.com

Comment: Remove virtual environment from release branch

LAST SYNC

Sync OK to 2e9cd23

Succeeded 5 minutes ago (Sat Jul 04 2026 23:27:22 GMT+0500)

Author: alex-n-permikai-alex-n-permikai@users.noreply.github.com

Comment: Remove virtual environment from release branch

devops-psu

devops-psu

service-devops

devops-psu

devops-psu-7498b8d985

devops-psu-6d6cf675b4

devops-psu-7498b8d985-b7xhf